

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 4 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCVC403 - STRUCTURAL MECHANICS - I

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following stress can also be known as hoop stress?
a. Axial stress b. Longitudinal stress
c. Fluid stress d. Circumferential stress
- 2 In a thin-walled cylinder subjected to internal pressure, the hoop/tangential stress is ----- times the longitudinal stress.
a. Half b. Twice
c. Three d. Equal
- 3 Conjugate-Beam method was developed by: -
a. Hooke b. Otto mohr
c. Charles E.greene d. H.muller-Breslau
- 4 The cylinder has a tendency to split up along due to circumferential stress.
a. Area b. Radius
c. Diameter d. Length
- 5 Pin joint is replaced by _____ in conjugate beam.
a. Roller b. pin
c. fixed support d. link
- 6 Oil tanks, steam boilers, gas pipes are examples of _____
a. Thick shells b. Thin cylinders
c. Hoop cylinders d. Longitudinal cylinders
- 7 The distribution factor is
a. ratio of stiffness of member and member
b. ratio of stiffness of near joint and far joint
c. ratio of stiffness of member and joint
d. ratio of stiffness of joint and member
- 8 Moment produced in a beam when Far End is Fixed is _____
a. $2EI/L$ b. $3EI/L$
c. $4EI/L$ d. $6EI/L$
- 9 Carryover Moment is defined as _____
a. The moment applied at one end to cause unit slope at the support.
b. The additional moment applied at one end to completely resist the rotation caused due to external loading.
c. The moment developed or induced at one end due to a moment at another end.
d. The moment applied at one end to cause unit slope at another end.

- 10 Deflection Equations method was developed by:-
 a) Mohr b) Bernoulli
 c) Maxwell d) Mohr and Manderla
- 11 While using slope deflection method, in which direction is the moment taken as positive?
 a) Clockwise b) Anti-clockwise
 c) Depends upon case d) Depends upon loading
- 12 If support B settles by 1mm downward, what is direction of rotation at point B?
 a) +ve
 b) -ve
 c) Can't say
 d) Depends upon loading at point A

Q2. Solve the following.

- A) Define the terms: 6
 a) Slope
 b) Deflection
 c) Displacement
 d) Elastic curve
 e) Radius of curvature
- B) Derive an expression for slope and deflection for simply supported beam with a UDL throughout span using double integration method? 6

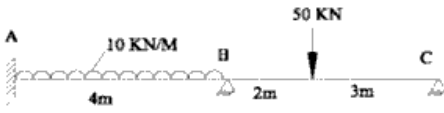
Q3. Solve the following.

- A) A cylindrical shell 3 m long has 1 m internal diameter and 15 mm metal thickness. Calculate the circumferential and longitudinal stress, if the shell is subjected to an internal pressure of 1.5 Mpa. Also calculate the changes in dimensions of the shell. Take $E = 200 \text{ GPa}$ and Poisson's ratio = 0.3 6
- B) Explain strain energy due to bending in detail. <https://www.batuonline.com> 6

Q4. Solve Any Two of the following.

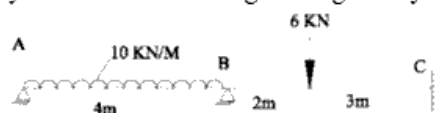
- A) State Castigliano's Theorem. Explain its assumptions and applications. Derive an expression for deflection at a point in a beam using strain energy due to bending. 6
- B) A simply supported beam of span L carries a point load P at mid-span. Using Castigliano's theorem, determine the deflection at mid-span and slope at the supports. Assume constant EI . 6
- C) A propped cantilever beam of span L is subjected to a uniformly distributed load w per unit length over the entire span. Determine the reaction at the prop and fixed end moment using compatibility conditions. 6

Q5. Solve Any Two of the following.

- A) Analyze the continuous beam given in figure by moment distribution method and draw the BMD and SFD 6
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- B) Define & explain the following Terms: 6
 a) Carry Over Moment b) Carry Over Factor c) Stiffness factor
- C) Explain the basic concept of the Moment Distribution Method. State the assumptions made in this method and discuss its advantages and limitations. 6

Q6. Solve Any Two of the following.

- A) Explain the assumptions made in Slope Deflection Method. 6
- B) Analyze the continuous beam given in figure 1 by slope deflection method and draw the BMD and SFD 6



- C) A continuous beam AB of span 6 m is fixed at A and hinged at B. The beam carries a uniformly distributed load of 20 kN/m over the entire span. Using the slope deflection method, determine the end moments at A and B.

6

***** End *****